

Package ‘msprog’

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compute_delta	<i>Default minimum clinically meaningful shift for different scales.</i>
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Description

Compute the minimum clinically meaningful score change as a function of the reference value for some widely used scales (EDSS, NHPT, T25FW, or SDMT), according to the most established rule for each of these outcomes.

Usage

```
compute_delta(baseline, outcome = "edss")
```

Arguments

baseline	Outcome value at baseline.
outcome	One of: • "edss" (Extended Disability Status Scale, default); • "nhpt" (Nine-Hole Peg Test); • "t25fw" (Timed 25-Foot Walk); • "sdt" (Symbol Digit Modalities Test).

Details

Default thresholds are meant to apply to all versions of each test (e.g., dominant or non-dominant hand for NHPT, best time or mean of two trials, etc.).

Value

Minimum clinically meaningful change from the provided baseline value. Specifically:

- EDSS: 1.5 if baseline=0, 1 if 0<baseline<=5.0, 0.5 if baseline>5.0
- NHPT and T25FW: 20% of baseline
- SDMT: either 4 points or 20% of baseline.

Examples

```
compute_delta(4.5) # default outcome is "edss"
compute_delta(55, outcome="sdt")
```

is_event	<i>Compare value to reference.</i>
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Description

Check if an outcome value determines a valid worsening, or improvement, or change, from a given reference value.

Usage

```
is_event(
  x,
  baseline,
  type,
  outcome = "edss",
  worsening = NULL,
  delta_fun = NULL,
  sub_threshold = FALSE
)
```

Arguments

x	Outcome value to test.
baseline	Outcome value at baseline.
type	One of: • "wors" (worsening); • "impr" (improvement); • "change" (any valid change).
outcome	One of: • "edss" (Extended Disability Status Scale, default); • "nhpt" (Nine-Hole Peg Test); • "t25fw" (Timed 25-Foot Walk); • "sdmt" (Symbol Digit Modalities Test); • "custom" (only accepted when specifying non-NULL worsening – and delta_fun as well, if sub_threshold=FALSE). Outcome type determines the direction of worsening (see worsening argument) and selects the default definition of clinically meaningful change given the reference value (using the built-in function compute_delta()). The latter can be replaced by a custom function using the delta_fun argument.
worsening	The direction of worsening ("increase" if higher values correspond to worse disease course, "decrease" otherwise). This argument is only used when outcome is set to "custom". Otherwise, worsening is automatically set to "increase" for EDSS, NHPT, T25FW, and to "decrease" for SDMT.

<code>delta_fun</code>	Custom function specifying the minimum clinically meaningful change in the outcome measure from the provided reference value. The function provided must take a numeric value (reference score) as input, and return a numeric value corresponding to the minimum relevant shift from baseline, see example below. If outcome is set to "custom", a custom delta function must be specified by the user. For other values of outcome, if no <code>delta_fun</code> is specified, the built-in function <code>compute_delta()</code> is used internally. The argument is ignored if <code>sub_threshold=TRUE</code> .
<code>sub_threshold</code>	If TRUE, any confirmed worsening, or improvement, or change in the outcome measure is valid, regardless of <code>delta_fun</code> .

Value

A boolean value specifying if a valid event was found.

Examples

```
is_event(x=4.5, baseline=4, type="wors", outcome="edss")
is_event(x=50, baseline=57, type="wors", outcome="sdmt")
is_event(x=3, baseline=3.5, type="impr", outcome="edss", sub_threshold=TRUE)
```

MSprog

Assess multiple sclerosis disability course from longitudinal data.

Description

Detect and characterise confirmed disability worsening (CDW) or improvement (CDI) events of an outcome measure (EDSS, NHPT, T25FW, or SDMT; or any custom outcome) based on repeated assessments through time (and on the dates of acute episodes, if any). The events are detected sequentially by scanning the outcome values in chronological order. Several qualitative and quantitative options are given as arguments that can be set by the user and reported as a complement to the results to ensure reproducibility.

Usage

```
MSprog(
  data,
  subj_col,
  value_col,
  date_col,
  outcome,
  relapse = NULL,
  rsubj_col = NULL,
  rdate_col = NULL,
  renddate_col = NULL,
  subjects = NULL,
  delta_fun = NULL,
  worsening = NULL,
  event = c("firstCDW", "firstCDI", "multiple", "firstPIRA", "firstRAW", "first"),
  RAW_PIRA = FALSE,
  baseline = c("fixed", "roving", "roving_impr", "roving_wors"),
```

```

proceed_from = c("firstconf", "event"),
sub_threshold_rebl = c("none", "change", "improvement", "worsening"),
bl_geq = FALSE,
relapse_rebl = FALSE,
skip_local_extrema = c("none", "strict", "all"),
validconf_col = NULL,
conf_days = 12 * 7,
conf_tol_days = c(7, 2 * 365.25),
require_sust_days = 0,
check_intermediate = T,
relapse_to_bl = 30,
relapse_to_event = 0,
relapse_to_conf = 30,
relapse_assoc = 90,
relapse_indep = NULL,
impute_last_visit = 0,
date_format = NULL,
include_dates = FALSE,
include_values = FALSE,
include_stable = TRUE,
verbose = 1
)

```

Arguments

data	Data frame containing longitudinal data, including: subject ID, outcome value, date of visit.
subj_col	Name of data column with subject ID.
value_col	Name of data column with outcome value.
date_col	Name of data column with date of visit.
outcome	Specifies the outcome type. Must be one of the following: • "edss" (Expanded Disability Status Scale) • "nhpt" (Nine-Hole Peg Test) • "t25fw" (Timed 25-Foot Walk) • "sdmt" (Symbol Digit Modalities Test) • "custom" (only accepted when specifying custom <code>delta_fun</code> and <code>worsening</code>). Outcome type triggers internal checks on value range, determines the direction of worsening (see <code>worsening</code> argument), and selects the default definition of clinically meaningful change given the reference value (using the built-in function <code>compute_delta()</code>). The latter can be replaced by a custom function using the <code>delta_fun</code> argument.
relapse	Optional data frame containing longitudinal data, including: subject ID and relapse date.
rsubj_col	Name of subject ID column for relapse data, if different from outcome data.
rdate_col	Name of onset date column for relapse data, if different from outcome data.
renddate_col	Name of end date column for relapse data (if applicable).
subjects	Subset of subjects (list of IDs). If none is specified, all subjects listed in data are included.

delta_fun	Custom function specifying the minimum clinically meaningful change in the outcome measure from the provided reference value. The function provided must take a numeric value (reference score) as input, and return a numeric value corresponding to the minimum shift from baseline, see example below. If outcome is set to "custom", a custom delta function must be specified by the user. For other values of outcome, if no delta_fun is specified, the built-in function <code>compute_delta()</code> is used internally.
worsening	The direction of worsening ("increase" if higher values correspond to worse disease course, "decrease" otherwise). The given value is only used when outcome is set to "custom". Otherwise, worsening is automatically set to "increase" if outcome is set to "edss", "nhpt", "t25fw", and to "decrease" if outcome is set to "sdmt".
event	Character string specifying which events to detect. Must be one of the following. • "firstCDW" (first CDW, default). • "firstCDI" (first CDI). • "multiple" (all confirmed events in chronological order). • "firstPIRA" (first progression independent of relapse activity, PIRA). • "firstRAW" (first relapse-associated worsening, RAW). • "first" (only the very first confirmed event – CDI or CDW).
RAW_PIRA	If TRUE, further classify CDW events based on their timing with respect to relapses (RAW, PIRA). The argument is ignored if event is set to "firstCDI" (not relevant) or to "firstRAW"/"firstPIRA" (always enabled).
baseline	Specifies the baseline scheme. Must be one of the following. • "fixed": first valid outcome value, default. • "roving": updated after each CDI or CDW event to the visit determined by <code>proceed_from</code>; suitable for a multiple-event setting (i.e., when event is set to "multiple") or when searching for a specific type of CDW (i.e., when event is set to "firstPIRA" or "firstRAW") – not recommended for randomised data. • "roving_impr": updated after every CDI (to the visit determined by <code>proceed_from</code>); suitable for a first-CDW setting to discard fluctuations around baseline – not recommended for multiple events, or for randomised data. • "roving_wors": updated after every CDW (to the visit determined by <code>proceed_from</code>); suitable when searching for a specific type of CDW (i.e., when event is set to "firstPIRA" or "firstRAW").
proceed_from	After detecting a confirmed disability event, continue searching: • from the next visit after the first qualifying confirmation visit if <code>proceed_from="firstconf"</code> • from the next visit after the event onset if <code>proceed_from="event"</code>. If baseline is set to "roving", "roving_impr", or "roving_wors", when re-baselining after a confirmed disability event, the baseline is moved to: • the first qualifying confirmation visit if <code>proceed_from="firstconf"</code> • the event visit if <code>proceed_from="event"</code>.
sub_threshold_rebl	This argument is only used if baseline is not set to "fixed". It controls whether and which "sub-threshold" events (i.e., <i>confirmed</i> changes in the outcome measure below the clinically meaningful threshold) can trigger a re-baseline. Must be one of the following.

- "none" (default): only use *clinically meaningful* confirmed changes for re-baseline.
- "change": any confirmed sub-threshold change can potentially trigger a re-baseline.
- "improvement": any confirmed sub-threshold improvement can potentially trigger a re-baseline.
- "worsening": any confirmed sub-threshold worsening can potentially trigger a re-baseline.

See `delta_fun` argument and `compute_delta()` function for more details.

<code>bl_geq</code>	This argument is only used if the baseline is moved. If TRUE, the new reference value must always be greater or equal than the previous one; when it is not, the old reference value is assigned to it [2].
<code>relapse_rebl</code>	If TRUE, re-baseline after every relapse [2].
<code>skip_local_extrema</code>	This argument is only used if the baseline is moved. It controls function behaviour in the presence of local minima or maxima. A visit i is a local minimum point for outcome x if $x[i+1] > x[i]$ and $x[i-1] > x[i]$. Local maxima are defined similarly. A visit i is a <i>strict</i> local minimum point for outcome x if $x[i+1] - x[i] \geq \text{delta_fun}(x[i])$; $x[i-1] - x[i] \geq \text{delta_fun}(x[i])$. Strict local maxima are defined similarly. When $x[i] = x[i-2]$, visit i is <i>not</i> considered a local extremum point even if the above conditions hold. This controls for cases where the outcome has an undulating course. The following argument values are accepted. • "none" (default): local extrema are always accepted as valid baseline values. • "strict": the baseline cannot be placed at a <i>strict</i> local minimum or maximum. • "all": the baseline cannot be placed at a local minimum or maximum.
<code>validconf_col</code>	Name of data column specifying which visits can (TRUE) or cannot (FALSE) be used as confirmation visits. The input data does not necessarily have to include such a column. If <code>validconf_col=NULL</code> , all visits are potentially used as confirmation visits.
<code>conf_days</code>	Period before confirmation (days). Can be a single value, or list-like of any length if considering multiple windows. If <code>length(conf_days) > 1</code> (e.g., <code>conf_days=c(12*7, 24*7)</code>), the function detects events confirmed at <i>either</i> time point (e.g., "confirmed at 12 <i>or</i> 24 weeks") with the relative tolerance (as per <code>conf_tol_days</code>).
<code>conf_tol_days</code>	Tolerance window for confirmation visit (days). Can be an integer (same lower and upper tolerance) or list-like of length 2 (different lower and upper tolerance). The right end of the interval (upper tolerance) can be set to Inf (confirmation window unbounded on the right – e.g., "confirmed over 12 <i>or more</i> weeks").
<code>require_sust_days</code>	Minimum number of days over which a confirmed change must be sustained (i.e., confirmed at <i>all</i> visits occurring in the specified period) to be retained as an event. Events sustained for the remainder of the follow-up period are always

retained regardless of follow-up duration. If `require_sust_days=Inf`, events are retained only when sustained for the remainder of the follow-up period.

(Warning: if `check_intermediate` is set to `FALSE`, sustained change will be established based *only on the end* of the specified period.)

`check_intermediate`

If `TRUE` (default), events are confirmed *over all intermediate visits* up to the confirmation visit.

If set to `FALSE` (not recommended in most cases, as it may discard meaningful fluctuations), events will be confirmed *only at* the specified confirmation visit (and *only at the end* of the period defined by `require_sust_days`, if any).

`relapse_to_bl`

Minimum distance (days) from the onset of a relapse for a visit to be used as baseline. Can be an integer (minimum distance from *last* relapse onset) or list-like of length 2 (minimum distance from *last* relapse onset, minimum distance from *next* relapse onset). Note that setting the distance to zero means keeping the baseline where it is regardless of surrounding relapses.

If relapse end dates are available (`renddate_col`), the minimum distance from last relapse onset is automatically set to the specific relapse duration, unless `relapse_to_bl` (or `relapse_to_bl[1]`) is zero (in which case relapse timing does not affect baseline placement).

If the designated baseline does not respect this constraint, the baseline is moved to the next available visit.

`relapse_to_event`

Minimum distance (days) from the onset of a relapse for an event to be considered as such. Can be an integer (minimum distance from *last* relapse onset) or list-like of length 2 (minimum distance from *last* relapse onset, minimum distance from *next* relapse onset). Note that setting the distance to zero means retaining the event regardless of surrounding relapses.

If relapse end dates are available (`renddate_col`), the minimum distance from last relapse onset is automatically set to the specific relapse duration, unless `relapse_to_event` (or `relapse_to_event[1]`) is zero (in which case relapse timing does not affect event validation).

`relapse_to_conf`

Minimum distance (days) from the onset of a relapse for a visit to be a valid confirmation visit. Can be an integer (minimum distance from *last* relapse onset) or list-like of length 2 (minimum distance from *last* relapse onset, minimum distance from *next* relapse onset). Note that setting the distance to zero means using any visit for confirmation regardless of surrounding relapses.

If relapse end dates are available (`renddate_col`), the minimum distance from last relapse onset is automatically set to the specific relapse duration, unless `relapse_to_conf` (or `relapse_to_conf[1]`) is zero (in which case relapse timing does not affect selection of confirmation visits).

`relapse_assoc`

Maximum distance (days) from the onset of a relapse for a CDW event to be classified as RAW. Can be an integer (maximum distance from *last* relapse onset) or list-like of length 2 (maximum distance from *last* relapse onset, maximum distance from *next* relapse onset). If relapse end dates are available (`renddate_col`), the maximum distance from last relapse is automatically set to the specific relapse duration. The argument is ignored if RAW events are not detected (e.g., if `event='firstCDW'` and `RAW_PIRA=FALSE`).

`relapse_indep`

Specifies relapse-free intervals for PIRA definition. Must be a named list `list(prec=list(p0, p1), event=list(e0, e1), conf=list(c0, c1))` specifying the intervals around (any subset of) three checkpoints:

1. a preceding visit, e.g., baseline or last visit before the worsening (p0 and p1)
2. the event onset (e0 and e1)
3. an eligible confirmation visit (c0 and c1).

The auxiliary function `relapse_indep_from_bounds()` can be used to organise interval bounds into a named list correctly, by calling:

```
relapse_indep_from_bounds(p0, p1, e0, e1, c0, c1)
```

See [relapse_indep_from_bounds\(\)](#) function docs for more details on how to define the intervals. If relapse end dates are available (`renddate_col`), it is possible to also define PIRA based on those by setting `use_end_dates=TRUE` in [relapse_indep_from_bounds\(\)](#). The argument is ignored if PIRA events are not detected (e.g., if `event='firstCDW'` and `RAW_PIRA=FALSE`).

`impute_last_visit`

Imputation probability for worsening events occurring at the last available visit (i.e., with no confirmation). Unconfirmed worsening events occurring at the last visit are never imputed if `impute_last_visit=0`; they are always imputed if `impute_last_visit=1`; they are imputed with probability p , $0 < p < 1$, if `impute_last_visit=p`. If a value $N > 1$ is passed, unconfirmed worsening events are imputed only if occurring within N days of follow-up (e.g., in case of early discontinuation).

`date_format`

Format of dates in the `date_col` and `rdate_col` columns of the input data. Can be specified as:

- Standard format for dates (e.g., "%d-%m-%Y"; see [strptime\(\)](#) docs for correct syntax).
- 'day' if dates in are given as "days from start" (the starting point can be different for each subject – e.g., days from randomisation in a clinical trial); negative values are accepted.

If not specified, function [as.Date\(\)](#) will try to infer it automatically.

`include_dates`

If TRUE, output\$results will include the dates of:

- event onset ('date' column)
- the current baseline ('bl_date' column)
- the last visit before event onset at a clinically meaningful score distance from it ('last_delta_date' column)
- the confirmation visit(s) ('conf<c>_date' column and, when relevant, 'PIRA_conf<c>_date' column for each c in `conf_days`)

`include_values`

If TRUE, output\$results will include the outcome value at:

- event onset ('value' column)
- the current baseline ('bl_value' column)
- the last visit before event onset at a clinically meaningful score distance from it ('last_delta_value' column)
- the confirmation visit(s) ('conf<c>_value' column and, when relevant, 'PIRA_conf<c>_value' column for each c in `conf_days`)

`include_stable`

If TRUE, subjects with no confirmed events are included in output\$results, with `time2event` = total follow up.

`verbose`

One of:

- 0 (print no info);
- 1 (print concise info, default);
- 2 (print extended info).

Value

An object of class `MSprogOutput` with the following attributes:

- `event_count`: a data frame containing event counts for each subject (and the event sequence in case of multiple events).
- `results`: a data frame with extended info on each event for all subjects.
- `settings`: a list containing all the arguments used to compute the output.
- `unconfirmed`: a data frame with info on unconfirmed events (initial change from baseline, but no confirmation) for all subjects.

For a detailed description of output data frames, see `?MSprogOutput`.

References

[1] Müller J, Cagol A, Lorscheider J, Tsagkas C, Benkert P, Yaldizli Ö, et al. Harmonizing definitions for progression independent of relapse activity in multiple sclerosis: A systematic review. *JAMA Neurol.* 2023;80:1232–45.

[2] Kappos L, Wolinsky JS, Giovannoni G, Arnold DL, Wang Q, Bernasconi C, et al. Contribution of relapse-independent progression vs relapse-associated worsening to overall confirmed disability accumulation in typical relapsing multiple sclerosis in a pooled analysis of 2 randomized clinical trials. *JAMA Neurol.* 2020;77:1132–40.

Examples

```
# 1. EDSS course
output <- MSprog(toydata_visits, subj_col='id', value_col='EDSS', date_col='date', outcome='edss',
  relapse=toydata_relapses, conf_days=12*7, conf_tol_days=30,
  event='multiple', baseline='roving', verbose=1)
print(output$results) # extended info on each event for all subjects
print(output$event_count) # event counts for each subject

# 2. SDMT course
output <- MSprog(toydata_visits, subj_col='id', value_col='SDMT', date_col='date', outcome='sdmt',
  relapse=toydata_relapses, conf_days=12*7, conf_tol_days=30,
  event='multiple', baseline='roving', verbose=1)
print(output$results) # extended info on each event for all subjects
print(output$event_count) # event counts for each subject

# 3. SDMT course, with a custom delta function
my_sdmtdelta <- function(reference_value) {min(c(reference_value/10, 3))}
output <- MSprog(toydata_visits, subj_col='id', value_col='SDMT', date_col='date', outcome='sdmt',
  delta_fun=my_sdmtdelta,
  relapse=toydata_relapses, conf_days=12*7, conf_tol_days=30,
  event='multiple', baseline='roving', verbose=1)
print(output$results) # extended info on each event for all subjects
print(output$event_count) # event counts for each subject
```

MSprogOutput

Objects of class "MSprogOutput"

Description

Objects of class `"MSprogOutput"` are returned by `MSprog()`. They contain the results of event detection and associated metadata.

Structure

Objects of class "MSprogOutput" are lists with the following elements:

- event_count** A data frame containing event counts for each subject (and the event sequence in case of multiple events).
- results** A data frame with extended info on each event for all subjects.
- settings** A list containing all the arguments used to compute the output.
- unconfirmed** A data frame with info on unconfirmed events (initial change from baseline, but no confirmation) for all subjects.

event_count data frame

event_sequence Sequence of detected events (if any), in chronological order (only displayed in multiple-event setting).

CDI Count of confirmed disability improvement (CDI) events.

CDW Count of confirmed disability worsening (CDW) events.

RAW Count of relapse-associated worsening (RAW) events.

PIRA Count of progression independent of relapse activity (PIRA) events.

Note: depending on computation settings, some of the above columns may not be included in the data frame (e.g., no CDI column if event='firstCDW' in `MSprog()`).

results data frame

<subj_col> Subject ID.

nevent Cumulative count of events for the subject.

event_type Type of event (CDI or CDW).

CDW_type Type of CDW (PIRA, RAW, undefined).

total_fu Total length of follow-up period in days.

time2event Number of days from start of follow-up to event onset.

bl2event Number of days from baseline to event onset.

date Date of event onset.

value Outcome value at event onset.

bl_date Date of baseline for the event.

bl_value Outcome value at baseline.

last_delta_date Date of last visit before event onset at a clinically meaningful score distance from it.

last_delta_value Last outcome value before event onset at a clinically meaningful score distance from it.

conf<conf_days>_date Date of <conf_days>-day confirmation visit for the event.

conf<conf_days>_value Outcome value at <conf_days>-day confirmation visit.

PIRA_conf<conf_days>_date When relevant, date of <conf_days>-day confirmation visit for PIRA (could be different from CDW confirmation visit if the PIRA definition includes confirmation-related constraints on relapses).

PIRA_conf<conf_days>_value When relevant, outcome value at <conf_days>-day confirmation visit for PIRA (could be different from CDW confirmation visit if the PIRA definition includes confirmation-related constraints on relapses).

sust_days Number of days for which the event was sustained.

sust_last Whether the event was sustained up to the end of follow-up.

Note: some of the columns may not be included, depending on computation settings (e.g., no date columns if include_dates=F in [MSprog\(\)](#)).

unconfirmed **data frame**

<subj_col> Subject ID.

date Date of initial outcome change.

value Outcome value at initial outcome change.

bl_date Date of baseline.

bl_value Outcome value at baseline.

closest_rel_before Distance in days from the last relapse before the outcome change (if any).

closest_rel_after Distance in days from the next relapse after the outcome change (if any).

Note: in first-event scenarios (e.g., event='firstCDW'), the event search stops as soon as the first confirmed event is found – in this case, the unconfirmed data frame only reports unconfirmed outcome changes occurring *before* detecting a confirmed event.

print.MSprogOutput	<i>Textual description of criteria used to assess disability course.</i>
--------------------	--

Description

print method for class "MSprogOutput".

Usage

```
## S3 method for class 'MSprogOutput'
print(x, ...)
```

Arguments

x	An object of class "MSprogOutput" (result of a call to MSprog()).
...	Optional arguments for print methods. They are ignored in this function.

Details

The method prints out (1) the package version, (2) a full list of function arguments, and (3) a short paragraph describing the full set of criteria used to obtain the output.

Examples

```
output <- MSprog(toydata_visits, "id", "EDSS", "date", "edss",
  relapse=toydata_relapses, conf_days=7*12, conf_tol_days=30,
  event="multiple", baseline="roving", verbose=2)
print(output) # textual description of parameters used to obtain output
```

relapse_indep_from_bounds

Define relapse-free intervals for PIRA definition.

Description

Organise the given interval bounds into a named list to be given as argument `relapse_indep` to function `MSprog()`. The relapse-free intervals may be anchored to (any subset of) the following three data-driven checkpoints.

1. "prec": a visit preceding the event: can be (i) the current baseline, (ii) the last visit before event onset, or (iii) the last visit before event onset with a clinically meaningful score distance from the event score (i.e., for an outcome x with "higher score" = "worse", the last visit i such that $x[\text{event}] - x[i] \geq \text{delta_fun}(x[i])$, and same for the confirmation visit).
2. "event": the disability worsening event onset.
3. "conf": an eligible confirmation visit.

Usage

```
relapse_indep_from_bounds(
  p0 = 0,
  p1 = 0,
  e0 = 0,
  e1 = 0,
  c0 = 0,
  c1 = 0,
  prec_type = "baseline",
  use_end_dates = F
)
```

Arguments

<code>p0</code>	Days before preceding visit (≥ 0).
<code>p1</code>	Days after preceding visit (≥ 0), or NULL.
<code>e0</code>	Days before event onset (≥ 0), or NULL.
<code>e1</code>	Days after event onset (≥ 0), or NULL.
<code>c0</code>	Days before confirmation (≥ 0), or NULL.
<code>c1</code>	Days after confirmation (≥ 0).
<code>prec_type</code>	Which visit to use as "preceding visit". Must be one of: • "baseline": the current baseline. • "last": the last visit before event onset. • "last_delta2": the last visit before event onset with a clinically meaningful score distance from the event score, i.e., for an outcome x with "higher score" = "worse", the last visit i where $x[\text{event}] - x[i] \geq \text{delta_fun}(x[i])$ and $x[\text{confirmation}] - x[i] \geq \text{delta_fun}(x[i])$.

use_end_dates If TRUE, only the right bounds e1 and c1 are used, as the left bounds will be defined by the onset-to-end interval of each relapse. This option is only relevant when relapse *end* dates are provided.

Details

If both ends of an interval are 0 (e.g., if both p0=0 and p1=0), the checkpoint is ignored. To merge two intervals together, set both the right end of the first interval and the left end of the second interval to NULL (e.g., "between baseline and event onset": p1=NULL and e0=NULL). Here are some examples:

- No relapses from 90dd before to 30dd after the event, and from 90dd before to 30dd after confirmation [1]:
relapse_indep_from_bounds(e0=90, e1=30, c0=90, c1=30).
- No relapses between baseline and confirmation (high-specificity definition from [1]):
relapse_indep_from_bounds(p0=0, p1=NULL, e0=NULL, e1=NULL, c0=NULL, c1=0).
- No relapses from baseline to 30dd after the event, and within confirmation+-30dd [2]:
relapse_indep_from_bounds(p0=0, p1=NULL, e0=NULL, e1=30, c0=30, c1=30).

Value

A named list to be given as argument relapse_indep to function `MSprog()`.

References

- [1] Müller J, Cagol A, Lorscheider J, Tsagkas C, Benkert P, Yaldizli Ö, et al. Harmonizing definitions for progression independent of relapse activity in multiple sclerosis: A systematic review. *JAMA Neurol.* 2023;80:1232–45.
- [2] Kappos L, Wolinsky JS, Giovannoni G, Arnold DL, Wang Q, Bernasconi C, et al. Contribution of relapse-independent progression vs relapse-associated worsening to overall confirmed disability accumulation in typical relapsing multiple sclerosis in a pooled analysis of 2 randomized clinical trials. *JAMA Neurol.* 2020;77:1132–40.

toydata_relapses	<i>Synthetic Relapse Data</i>
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Description

Artificially generated relapse dates for some example patients in `toydata_visits` to illustrate the use of the package.

Usage

```
data(toydata_relapses)
```

Format

An object of class `data.frame`.

id Subject ID.

date The relapse date.

References

This data set was artificially created for the msprog package.

Examples

```
data(toydata_relapses)
head(toydata_relapses)
```

toydata_visits

Synthetic Longitudinal EDSS and SDMT Data

Description

Artificially generated toy data set providing Extended Disability Status Scale (EDSS) and Symbol Digit Modalities Test (SDMT) scores in a small cohort of example patients to illustrate the use of the package.

Usage

```
data(toydata_visits)
```

Format

An object of class `data.frame`.

id Subject ID.

date The visit date.

EDSS A value between 0 and 10.

SDMT A value between 0 and 110.

References

This data set was artificially created for the msprog package.

Examples

```
head(toydata_visits)
```

value_milestone	<i>Time to disability milestone.</i>
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Description

Scan the visits in chronological order to detect the first outcome value reaching or exceeding a specified disability milestone (e.g., EDSS $\geq$ 6), *with confirmation*.

Usage

```
value_milestone(
  data,
  milestone,
  subj_col,
  value_col,
  date_col,
  outcome,
  worsening = NULL,
  relapse = NULL,
  rsubj_col = NULL,
  rdate_col = NULL,
  validconf_col = NULL,
  conf_days = 12 * 7,
  conf_tol_days = c(7, 2 * 365.25),
  require_sust_days = 0,
  relapse_to_event = 0,
  relapse_to_conf = 30,
  impute_last_visit = 0,
  date_format = NULL,
  verbose = 0
)
```

Arguments

data	Data frame containing longitudinal data, including: subject ID, outcome value, date of visit.
milestone	Disability milestone (outcome value to check data against).
subj_col	Name of data column with subject ID.
value_col	Name of data column with outcome value.
date_col	Name of data column with date of visit.
outcome	Specifies the outcome type. Must be one of the following: • "edss" (Expanded Disability Status Scale) • "nhpt" (Nine-Hole Peg Test) • "t25fw" (Timed 25-Foot Walk) • "sdmt" (Symbol Digit Modalities Test) • "custom" (only accepted when specifying argument worsening). Outcome type triggers internal checks on value range and determines the direction of worsening (see worsening argument).

worsening	The direction of worsening ("increase" if higher values correspond to worse disease course, "decrease" otherwise). The given value is only used when outcome is set to "custom". Otherwise, worsening is automatically set to "increase" if outcome is set to "edss", "nhpt", "t25fw", and to "decrease" if outcome is set to "sdmt".
relapse	Optional data frame containing longitudinal data, including: subject ID and relapse date.
rsubj_col	Name of subject column for relapse data, if different from outcome data.
rdate_col	Name of date column for relapse data, if different from outcome data.
validconf_col	Name of data column specifying which visits can (TRUE) or cannot (FALSE) be used as confirmation visits. The input data does not necessarily have to include such a column. If validconf_col=NULL, all visits are potentially used as confirmation visits.
conf_days	Period before confirmation (days). Can be a single value, or list-like of any length if considering multiple windows. If length(conf_days) > 1 (e.g., conf_days=c(12*7, 24*7)), the function retains milestones confirmed at <i>either</i> time point (e.g., "confirmed at 12 <i>or</i> 24 weeks") with the relative tolerance (as per conf_tol_days).
conf_tol_days	Tolerance window for confirmation visit (days); can be an integer (same lower and upper tolerance) or list-like of length 2 (different lower and upper tolerance). The right end of the interval can be set to Inf (confirmation window unbounded on the right – e.g., "confirmed over 12 <i>or more</i> weeks").
require_sust_days	Minimum number of days over which the milestone must be sustained (i.e., confirmed at <i>all</i> visits occurring in the specified period). If the milestone is sustained for the remainder of the follow-up period, it is considered reached regardless of follow-up duration. If require_sust_days=Inf, values are retained only when sustained for the remainder of the follow-up period.
relapse_to_event	Minimum distance (days) from the onset of a relapse for the milestone to be considered reached. Can be an integer (minimum distance from <i>last</i> relapse onset) or list-like of length 2 (minimum distance from <i>last</i> relapse onset, minimum distance from <i>next</i> relapse onset). Note that setting the distance to zero means retaining the event regardless of surrounding relapses.
relapse_to_conf	Minimum distance (days) from the onset of a relapse for a visit to be a valid confirmation visit. Can be an integer (minimum distance from <i>last</i> relapse onset) or list-like of length 2 (minimum distance from <i>last</i> relapse onset, minimum distance from <i>next</i> relapse onset). Note that setting the distance to zero means using any visit for confirmation regardless of surrounding relapses.
impute_last_visit	Imputation probability when the milestone is reached at the last available visit (i.e., with no confirmation). Unconfirmed values exceeding the milestone at the last visit are never imputed if impute_last_visit=0; they are always imputed if impute_last_visit=1; they are imputed with probability p, 0<p<1, if impute_last_visit=p. If a value N>1 is passed, unconfirmed values exceeding the milestone are imputed only if occurring within N days of follow-up (e.g., in case of early discontinuation).
date_format	Format of dates in the input data. Can be:

- "day" if dates are given as "days from start" (the starting point can be different for each subject – e.g., days from randomisation in a clinical trial); negative values are accepted.
- Standard format for dates (e.g., "%d-%m-%Y"; see [strptime\(\)](#) docs for correct syntax).

If not specified, function [as.Date\(\)](#) will try to infer it automatically.

verbose

One of:

- 0 (print no info)
- 1 (print concise info, default)
- 2 (print extended info).

Details

- "Reaching or exceeding" means either $\text{value} \geq \text{milestone}$ or $\text{value} \leq \text{milestone}$, depending on the direction of worsening (see arguments `outcome` and `worsening`).
- An event is only considered "observed" if **confirmed**, i.e., if all values *up to* the confirmation visit reach or exceed the milestone.

Value

A data frame containing the following columns:

- `<date_col>`: the date of first reaching or exceeding the milestone with confirmation (or last date of follow-up if milestone is not reached or not confirmed)
- `<value_col>`: the first value reaching or exceeding the milestone with confirmation, if present, otherwise no value
- `"time2event"`: the time taken to reach or exceed the milestone (or total follow-up length if milestone is not reached or not confirmed)
- `"observed"`: whether the milestone was reached with confirmation (1) or not (0).

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